

Q1.2)

$$1^2 + 2^2 + 3^2 + \dots + (2n)^2 = \frac{n(2n+1)(4n+1)}{3}$$

$$1^2 + 2^2 = 1 + 4 = 5$$

$$\frac{1(2(1)+1)(4(1)+1)}{3}$$

$$\frac{(3)(5)}{3} = 5 \quad \checkmark$$

hyp

Assume $n = k$

$$+(2k)^2 = \frac{k(2k+1)(4k+1)}{3}$$

Step $n = k+1$

$$+(2(k+1))^2 = \frac{(k+1)(2(k+1)+1)(4(k+1)+1)}{3}$$

$$(2k+2)^2 = \frac{k+1(2k+3)(4k+5)}{3}$$

$$\frac{k(2k+1)(4k+1)}{3} + (2k+1)^2 + (2k+2)^2$$

$$\frac{k(2k+1)(4k+1) + 3(2k+1)^2 + 3(2k+2)^2}{3}$$

FOIL

$$(2k^2 + k)(4k+1)$$

$$\frac{8k^3 + 2k^2 + 4k^2 + k + 3(2k+1)^2 + 3(2k+2)^2}{3}$$

$$(2k+1)(2k+1)$$

$$8k^3 + 6k^2 + k + 3(4k^2 + 4k + 1) + 3(2k+2)^2$$

$$8k^3 + 6k^2 + k + 12k^2 + 12k + 3 + 3(4k^2 + 8k + 4)$$

$$\frac{8k^3 + 18k^2 + 13k + 3 + 12k^2 + 24k + 12}{3}$$

$$\frac{8k^3 + 30k^2 + 13k + 15}{3}$$

$$\frac{8k^3 + 30k^2 + 37k + 15}{3}$$

$$(8k^3 + 8k^2) + (22k^2 + 22k) + (15k + 15)$$

$$8k^2(k+1) + 22k(k+1) + 15(k+1)$$

$$(k+1)(8k^2 + 22k + 15)$$

$$(k+1)(8k^2 + 12k + 10k + 15) \quad \begin{matrix} 8 \times 15 = 120 \\ 12 \times 10 = 120 \\ 12 + 10 = 22 \end{matrix}$$

$$\downarrow \quad 4k(2k+3) + 5(2k+3)$$

$$(k+1)(2k+3)(4k+5)$$

$$\frac{\text{Left} \quad 3}{(k+1)(2k+3)(4k+5)} \quad \begin{matrix} k+1 \\ \checkmark \text{ It is True} \end{matrix}$$

$$\frac{\text{Right}}{(k+1)(2(k+1)+1)(4(k+1)+1)} \quad 3$$

Both of the last 2 are the same
so for $k+1$ all $n \geq 1$ are true by
Proof by Induction

$$\text{Right} = \text{Left for } k+1 \text{ and } n$$

Q1.3)

if you have 3 coins Assume 1 is lighter then you can split the total coins and weigh them as long as both sides have equal coins if not you must weigh with left overs till you find the coin that is not like the rest

(A, B, C)

if $A > C$

Then $\checkmark$ A is the counterfeit
Then C vs B make sure they are the same

if $C > A$

C is the counterfeit

if $C = A$

B is the counterfeit

If larger Just split every time

A, B, C, D, E, F

(A, B, C) > (D, E, F)

↖ counterfeit on this side binary down

$3^1 \quad n=0$

$$3^1 = 3^0 = 1 \text{ maybe min}$$

hyp

if $3^0 = 1$ the $n = \text{true}$ so

$n+1 = \text{true}$

Step

$$n+1 = k \quad \checkmark$$

$$3^k = 3^1 = 3 \text{ True}$$